



Data & Analytics for AI in HR

SQL | Python | Power BI | DAX | Power Query | AI | Automation

Purpose

A practical AI in HR Analytics Lab demonstrating how HR data moves from raw records to validated metrics, executive analytics, AI-assisted insight and automated decision workflows. The positioning is deliberately HR analytics practitioner / transformation lead - not data scientist, software engineer or HRIS implementer.

Architecture

HR Data -> SQL / Python -> Data Quality & Validated Metrics -> Power BI / DAX -> AI-Assisted Insight -> Human HR Decision -> Workflow Automation

Portfolio evidence

This GitHub project includes a synthetic HR dataset, executable-style SQL examples, a Python/pandas analytics script, Power BI DAX measures and an analytical workbook with executive, skills and recruitment views. No real employee data is used.

01 | SQL for HR Analytics

Demonstrates querying, grouping and validating HR data for workforce, skills, mobility, recruitment and attrition analysis.

Practical evidence

- Workforce headcount, tenure, performance and exit rates by department.
- Skills coverage, average proficiency and capability gaps.
- Internal mobility rates by organisational level.
- Recruitment source and average time-to-fill.
- Attrition pattern analysis by department and tenure band - for aggregate investigation, not individual prediction.

GitHub artifact

``hr_analytics_queries.sql``

02 | Python for HR Analytics & AI Preparation

Demonstrates pandas-based data validation, analysis and preparation of governed aggregate context for AI.

Practical evidence

- Validate employee IDs and rating/skill ranges.
- Calculate workforce KPIs and departmental metrics.
- Create skills intelligence and proficiency-gap analysis.
- Prepare aggregate, validated metrics for an approved enterprise LLM.
- Keep AI downstream of metric validation rather than asking an LLM to calculate authoritative HR KPIs.

GitHub artifact

``hr_analytics_python_example.py``

03 | Power BI, DAX & Power Query

Demonstrates the executive decision-intelligence layer used after data preparation and validation.

Practical evidence

- DAX measures for headcount, exits, exit rate, internal mobility, time-to-fill, learning and skills.
- Power Query concept for repeatable cleaning and transformation.
- Executive workforce dashboard with departmental drill-down.
- Skills/capability view and recruitment source/time-to-fill view.
- Metrics remain traceable to validated source fields and defined business rules.

GitHub artifact

`power\_bi\_dax\_measures.txt` + `HumanEvoX\_AI\_in\_HR\_Analytics\_Lab.xlsx`

04 | AI Insight & HR Automation Layer

Shows where AI and automation add value after the analytics layer is trusted.

Practical evidence

- LLM summarises validated aggregate trends and drafts executive commentary.
- AI classifies governed HR text or skills terminology for analyst validation.
- Thresholds can trigger Power Automate workflows and analyst review.
- Approved briefing can be distributed after human validation.
- No autonomous employment decision, individual flight-risk score or unvalidated AI metric.

GitHub artifact

Design pattern: KPI threshold -> validated metric -> AI draft -> analyst check -> approved action / briefing.